

UNIT – 2

CYBER CRIME : Mobile and Wireless Devices-Introduction, Proliferation of Mobile and Wireless Devices, Trends in Mobility, Credit Card Frauds in Mobile and Wireless Computing Era, Security Challenges Posed by Mobile Devices, Registry Settings for Mobile Devices, Authentication Service Security, Attacks on Mobile/Cell Phones, Mobile Devices: Security Implications for organizations, Organizational Measures for Handling Mobile, Organizational Security Policies and Measures in Mobile Computing Era.

1 INTRODUCTION TO MOBILE AND WIRELESS DEVICES

1.1 What Are Mobile Devices?

Mobile devices are portable computing devices that can process, store, and transmit data wirelessly.

Examples:

- Smartphones
- Tablets
- Laptops
- Smartwatches
- IoT devices (CCTV, smart bulbs)

1.2 What Is Wireless Communication?

Wireless communication means sending data without cables using radio waves.

Technologies:

- Wi-Fi (802.11)
- Bluetooth
- NFC
- 3G / 4G / 5G cellular network

1.3 Why Mobile Devices Are Security Concern?

Earlier:

- Data stored in desktop computers (fixed location)

Now:

- Data in pockets (mobile phones)

- Banking, emails, office login, cloud access

If mobile compromised → full digital identity compromised.

Example:

If attacker gains access to your mobile:

- Reads OTP
- Access bank app
- Reset passwords
- Steal money

2 PROLIFERATION OF MOBILE AND WIRELESS DEVICES

2.1 Meaning of Proliferation

Proliferation = Rapid increase in number.

India Example:

- UPI revolution
- Affordable smartphones
- Jio data revolution

2.2 Why Rapid Growth Happened?

1. Cheap internet
2. Digital India movement
3. Work From Home
4. Cloud computing
5. Online education

2.3 Security Impact of Proliferation

Engineering Concept: Attack Surface

Attack surface = Total possible entry points.

Earlier:

1 desktop system → 1 entry point

Now:

- Laptop
- Mobile
- Tablet
- Smartwatch
- Home router

More devices → More vulnerabilities → More risk.

3 TRENDS IN MOBILITY

3.1 BYOD (Bring Your Own Device)

Employees use personal mobile for office work.

Risk:

- Personal apps mix with company data.
- Malware in personal app can access office email.

3.2 Remote Work Culture

Employees connect to company server from home.

Risk:

- Weak home WiFi password
- Shared network
- No firewall

3.3 Cloud-Based Apps

Mobile apps store data in cloud.

Risk:

- Account takeover
- API vulnerabilities
- Weak authentication

3.4 Mobile Payment Systems

UPI, Wallet, NFC payments.

Risk:

- QR code scam
- Fake payment request
- Screen sharing fraud



CREDIT CARD FRAUDS IN MOBILE AND WIRELESS ERA

4.1 How Credit Card Fraud Happens

Method 1: Phishing App

Fake banking app collects card details.

Method 2: Smishing

SMS with fake link.

Method 3: Malware

Trojan records keystrokes.

Method 4: Public WiFi Interception

Man-in-the-middle attack.

4.2 Technical Flow of Fraud

1. User installs malicious app.
2. App asks permissions.
3. User enters card details.
4. Data sent to attacker server.
5. Attacker performs transaction.

4.3 Modern Fraud Techniques

- SIM swap fraud
- Screen sharing app misuse
- Fake customer care numbers
- OTP forwarding malware

4.4 Prevention Measures

- Install apps only from Play Store/App Store
- Enable transaction alerts
- Disable screen sharing
- Never share OTP
- Use virtual cards

5 SECURITY CHALLENGES POSED BY MOBILE DEVICES

5.1 Physical Loss or Theft

Mobile is portable → easily lost.

Risk:

- Data exposure
- Banking app misuse

Solution:

- Device encryption
- Remote wipe
- Biometric lock

5.2 Malware Infection

APK from unknown source.

Types:

- Spyware
- Trojan
- Ransomware

5.3 Weak Authentication

PIN 1234 or pattern lock.

Solution:

- Strong password
- Biometric + PIN
- MFA

5.4 Unsecured Wireless Communication

Public WiFi risk.

Attack:

- Packet sniffing
- Evil twin attack

5.5 Data Leakage via Apps

Apps requesting:

- Contacts
- Camera
- Microphone
- Storage

If compromised → data leak.

6 REGISTRY SETTINGS FOR MOBILE DEVICES

Registry = System configuration settings controlling behavior.

In mobile devices (Android/iOS), it includes:

- Security policies
- Permission management
- Device management settings

Important Security Settings

1. Disable unknown app installation
2. Enable encryption
3. Set auto-lock timer
4. Enforce password complexity
5. Disable USB debugging
6. Restrict Bluetooth visibility

Engineering analogy:

Like configuring firewall rules in operating system.

7 AUTHENTICATION SERVICE SECURITY

Authentication = Verifying identity before access.

7.1 Types of Authentication Factors

1. Something You Know

Password, PIN

2. Something You Have

OTP, token, SIM

3. Something You Are

Fingerprint, Face ID

7.2 Multi-Factor Authentication (MFA)

Using two or more factors.

Example:

UPI PIN + OTP.

7.3 Threats to Authentication

- Brute force attack
- SIM swap
- OTP interception malware
- Shoulder surfing
- Phishing login pages

8 ATTACKS ON MOBILE / CELL PHONES

8.1 Malware Attacks

Types:

- Trojan
- Spyware
- Ransomware
- Adware

Example:

Fake cracked game app.

8.2 Bluetooth Attacks

- Bluejacking (sending unwanted messages)
- Bluesnarfing (stealing data)

8.3 WiFi Attacks

- Man-in-the-middle
- Rogue access point
- Packet sniffing

8.4 SIM Swap Attack

Attacker convinces telecom provider to issue duplicate SIM.

Result:

Receives OTP → empties bank account.

8.5 Mobile Ransomware

Encrypts phone data and demands ransom.

9 MOBILE DEVICES: SECURITY IMPLICATIONS FOR ORGANIZATIONS

Organizations face major risks:

9.1 Data Leakage

Employee storing company files in personal phone.

9.2 Insider Threat

Employee intentionally leaking data.

9.3 Unauthorized Access

Stolen device → attacker accesses company email.

9.4 Shadow IT

Employee installing unauthorized cloud apps.

10 ORGANIZATIONAL MEASURES FOR HANDLING MOBILE

10.1 Mobile Device Management (MDM)

Software used to:

- Monitor devices
- Enforce security policies
- Remote wipe

10.2 Containerization

Separate work and personal data.

Two isolated environments.

10.3 Remote Wipe Capability

Delete company data if device lost.

10.4 VPN Enforcement

Force encrypted connection to company servers.

10.5 Encryption Enforcement

Full device encryption mandatory.

1 1 ORGANIZATIONAL SECURITY POLICIES IN MOBILE ERA

11.1 BYOD Policy

Defines:

- Allowed apps
- Security requirements
- Monitoring rights

11.2 Acceptable Use Policy

Defines what employees can and cannot do.

11.3 Incident Response Policy

Steps after mobile breach:

1. Report
2. Isolate device
3. Remote wipe
4. Investigate

11.4 Data Classification Policy

Data categorized as:

- Public
- Internal
- Confidential
- Restricted

Access based on classification.

1 BYOD vs Corporate Owned Devices

Basis	BYOD (Bring Your Own Device)	Corporate Owned Device
Ownership	Employee	Organization
Control	Limited control	Full administrative control
Security Risk	High	Lower
Monitoring	Restricted (privacy issue)	Fully allowed
Data Separation	Difficult	Easy to enforce
Example	Employee using personal phone for office mail	Company-issued secure phone

2 Single Factor vs Multi-Factor Authentication

Basis	Single-Factor Authentication	Multi-Factor Authentication
Security Level	Low	High
Factors Used	One (Password only)	Two or more
Risk	Easy to hack	Harder to compromise
Example	Only password	Password + OTP
Usage	Basic login systems	Banking apps

3 WiFi vs Bluetooth (Security Perspective)

Basis	WiFi	Bluetooth
Range	Long (50–100 meters)	Short (10 meters)
Usage	Internet access	Device-to-device
Common Attack	Evil twin, MITM	Bluesnarfing
Encryption	WPA2/WPA3	Pairing-based encryption
Risk Level	High (public WiFi)	Medium

4 Phishing vs Smishing vs Vishing

Basis	Phishing	Smishing	Vishing
Medium	Email	SMS	Voice call
Target	Mass users	Mobile users	Bank customers
Example	Fake bank mail	Fake OTP SMS	Fake bank officer call
Risk	Credential theft	OTP theft	Direct fraud

5 Centralized Botnet vs Peer-to-Peer Botnet

Basis	Centralized Botnet	P2P Botnet
Control	Single command server	Distributed nodes
Detection	Easier	Harder
Shutdown	Easy (remove server)	Difficult
Structure	Star topology	Decentralized
Example	Basic IRC botnet	Advanced botnets

6 Mobile Malware vs Desktop Malware

Basis	Mobile Malware	Desktop Malware
Target	Android/iOS	Windows/Linux
Spread	Fake apps, APK	Email attachment
Permissions	App-based access	System-level access
Detection	Hard (less antivirus)	Easier (more tools)
Example	Banking trojan	Ransomware

7 Mobile Device Management (MDM) vs Containerization

Basis	MDM	Containerization
Control	Full device control	Only work data controlled
Privacy	Less privacy for user	More privacy
Data Separation	Entire device managed	Work and personal separated

Use Case	Corporate devices	BYOD
Example	Company remotely wiping phone	Work profile in Android

8 SIM Swap Attack vs Phishing Attack

Basis	SIM Swap	Phishing
Method	Duplicate SIM issued	Fake website/email
Target	OTP system	Login credentials
Risk	Bank account takeover	Account compromise
Detection	Difficult	Sometimes detectable
Medium	Telecom manipulation	Internet

9 Public WiFi vs Secure Private Network

Basis	Public WiFi	Private Network
Security	Weak	Strong
Encryption	Often absent	WPA2/WPA3
Risk	MITM, sniffing	Low
Usage	Airports, cafes	Home/Office
Recommendation	Use VPN	Generally safe

10 Device Encryption vs App-Level Encryption

Basis	Device Encryption	App-Level Encryption
Scope	Entire device data	Specific app data
Protection Level	High	Medium
Example	Android full disk encryption	WhatsApp end-to-end
Use Case	Lost phone protection	Secure messaging

1 Remote Wipe vs Factory Reset

Basis	Remote Wipe	Factory Reset
Trigger	Organization/admin	User manually
Purpose	Remove company data	Reset device
Control	IT department	User

Use Case	Lost corporate device	Selling phone
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1 2 Mobile Payment Fraud vs Credit Card Skimming

Basis	Mobile Payment Fraud	Card Skimming
Medium	App-based	ATM/POS device
Method	Fake apps, OTP theft	Card cloning
Location	Online	Physical
Technology Used	Malware	Skimming device

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